Online Bookstore Analysis using SQL

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Project Overview:

Project Type: Self

Platform: MySQL Workbench

Objectives:

This project is a SQL-based data analysis project built on an Online Bookstore Dataset. The goal of this project is to analyze book inventory, customer behavior, orders, sales performance, and revenue trends using SQL queries ranging from basic to advanced level.

Tools & Skills Used:

- SQL
- MySQL / Relational Database Concepts
- Joins
- Aggregate Functions
- Group BY / Order By
- Filtering and Sorting
- Business Actionable Insight Generation from Raw Data

❖ Dataset Information:

I. Books Table (.csv file)

- **Columns**: Book ID, Title, Author, Genre, Published Year, Price, Stock
- **Purpose:** Stores book details such as title, author, genre, price, and stock, and supports inventory and product analysis.

I. Customer Table (.csv file)

- **Columns:** Customer ID, Name, City, Country
- **Purpose:** Stores customer information and helps analyze customer behavior, location, and spending patterns.

I. Orders Table (.csv file)

- **Columns:** Order ID, Customer ID, Book ID, Order Date, Quantity, Total Amount
- **Purpose:** Stores transaction records and is used to analyze sales, revenue, quantity sold, and purchase trends.

❖ Dataset Overview:

The dataset used in this project is sourced from Kaggle, titled "Online Bookstore Analysis." It contains bookstore-related transactional and master data that is well-suited for SQL practice and analysis. The dataset primarily focuses on key business areas such as inventory management, sales performance, revenue insights, stock availability, and customer behavior.

❖ Project Objectives:

The main objectives of this project are:

- **Analyze book inventory and stock availability**
- **Explore customer purchase behavior**
- **Track sales and revenue performance**
- **Identify top-performing books and authors**
- **Practice SQL queries from beginner to advanced level**
- **Generate meaningful business insights from structured data**

- **Business Problem**:

The online bookstore wants to improve decision-making by using SQL-driven insights into customer behavior, sales performance, and inventory status.

- **Customer Behavior :**
  - Identify repeat and high-spending customers.
  - Analyze customer distribution by city and country.
  - Find customers who placed multiple orders.

- **Sales Performance :**
  - Track best-selling books and popular genres.
  - Measure total revenue generated from orders.
  - Identify the most frequently ordered books and top-performing authors.

- **Inventory Status :**
  - Detect low-stock and out-of-stock books.
  - Monitor stock remaining after fulfilling customer orders.
  - Evaluate book availability across different genres.

- **Goal:**

Enable the business to uncover buying patterns, optimize inventory management, improve sales tracking, and focus on the most profitable areas of the bookstore.

Customer Insights

Query 1: List customers who have placed at least 2 orders.

```
SELECT C.CustomerID,
        C.Name,
        COUNT(O.OrderID) AS Total_Ordered
FROM Customers AS C
INNER JOIN Orders AS O
ON C.CustomerID = O.CustomerID
GROUP BY C.CustomerID
HAVING COUNT(O.OrderID) > 1
ORDER BY C.CustomerID;
```

Result Grid | Filter Rows: | Export:

CustomerID	Name	Total_Ordered
2	Crystal Clements	2
6	Stephen Vasquez	2
8	Matthew Johnson	2
13	Kristine Kim	2
14	John Wood	2
15	Vanessa Gaines	2
16	Stacey Flores	3
21	Edgar Frost	2
22	Stacey Adams	3
23	Hannah Drake	2
24	Christina Mitchell	2
37	Ryan Lara	2
47	Debra Park	3
54	Rachel Jones	2

Customer Insights

Query 2: List the cities where customers who spent over $30 are located.

```
SELECT ROW_NUMBER() OVER() AS 'Sr.No.',
        C.City,
        Total_Amount
FROM Customers AS C
INNER JOIN Orders AS O
ON C.CustomerID = O.CustomerID
WHERE Total_Amount > 30
ORDER BY Total_amount DESC;
```

Sr.No.	City	Total_Amount
1	Smithborough	491.50
2	Thomaschester	489.60
3	Fieldsland	489.60
4	Mendezburgh	486.70
5	Melissaside	480.30
6	Freemanland	469.70
7	Bakerton	469.30
8	West Kimberlyhaven	466.60
9	Elizabethshire	465.40
10	Lake Charleshaven	459.10
11	Buckbury	452.00
12	Crystalborough	449.01
13	Erikaberg	446.31
14	Lake Tyler	445.50
15	Jenniferfurt	442.80

Customer Insights

Query 3: Find the customer who spent the most on orders

```
SELECT C.CustomerID,
       C.Name,
       SUM(O.Total_Amount) AS Total_Spent
FROM Customers AS C
INNER JOIN Orders AS O
ON C.CustomerID = O.CustomerID
GROUP BY C.CustomerID,C.Name
ORDER BY Total_Spent DESC
LIMIT 1;
```

CustomerID	Name	Total_Spent
457	Kim Turner	1398.90

Sales Insights

Query 4: Find the most frequently ordered book.

```
SELECT O.BookID,
        B.Title,
        B.Author,
        SUM(O.Quantity) AS total_ordered
FROM Books AS B
INNER JOIN Orders AS O
ON B.BookID = O.BookID
GROUP BY B.BookID,B.Title,B.Author
ORDER BY total_Ordered DESC,O.BookID ASC
LIMIT 5;
```

BookID	Title	Author	total_ordered
73	Realigned multi-tasking installation	Patrick Contreras	28
31	Implemented encompassing conglomeration	Melissa Taylor	27
157	Advanced didactic time-frame	Thomas Trujillo	24
273	Devolved zero administration process improvem...	Emily James	24
120	Integrated secondary access	Sheena Harris	23

Sales Insights

Query 5: Show the top 3 most expensive books of 'Fantasy' Genre.

```
SELECT BookID,
        Title,
        Genre,
        Price
FROM Books
WHERE Genre = 'Fantasy'
ORDER BY Price DESC
LIMIT 3;
```

BookID	Title	Genre	Price
240	Stand-alone content-based hub	Fantasy	49.90
462	Innovative 3rdgeneration database	Fantasy	49.23
238	Optimized even-keeled analyzer	Fantasy	48.97
NULL	NULL	NULL	NULL

Sales Insights

Query 6: Calculate the total revenue generated from all orders.

```
SELECT SUM(Total_Amount) AS Total_Revenue
FROM Orders;
```

	Total_Revenue
▶	75443.27

Sales Insights

Query 7: Retrieve all orders where the total amount exceeds $20.

```
SELECT *
FROM Orders
WHERE Total_amount > 20.00;
```

OrderID	CustomerID	BookID	Order_Date	Quantity	Total_amount
1	84	169	2023-05-26	8	188.56
2	137	301	2023-01-23	10	216.60
3	216	261	2024-05-27	6	85.50
4	433	343	2023-11-25	7	301.21
5	14	431	2023-07-26	7	136.36
6	439	119	2024-10-11	5	249.40
7	195	467	2023-10-23	6	82.92
8	32	159	2024-05-07	4	144.84
9	109	407	2024-01-04	9	379.71
10	94	94 2	2024-07-09	4	123.00
11	131	206	2023-10-16	1	38.01
12	454	3	2024-06-17	2	31.50
13	420	180	2023-06-08	5	125.45
14	454	319	2023-08-24	2	85.22
15	127	479	2023-01-10	6	229.62

Inventory Insights

Query 8: Retrieve the total quantity of books sold by each author.

```
SELECT B.Author,
        SUM(O.Quantity) AS Total_Quantity
FROM Books AS B
INNER JOIN Orders AS O
ON O.BookID = B.BookID
GROUP BY B.Author
ORDER BY Total_Quantity DESC;
```

Author	Total_Quantity
Patrick Contreras	28
Melissa Taylor	27
Emily James	24
Thomas Trujillo	24
Erica Parker	23
Sheena Harris	23
Valerie Moore	23
Ellen Doyle	23
Rachel Gibbs	22
Amanda Wilson	22
Tonya Saunders	21
Kristi Phillips	21
Anna Roberts	21
Michael Hill	20
Alexander Wallace	20

Inventory Insights

Query 9: Find the book with the lowest stock.

```
SELECT BookID,
        Title,
        Author,
        Stock
FROM Books
ORDER BY Stock ASC
LIMIT 5;
```

BookID	Title	Author	Stock
378	Future-proofed heuristic function	Samantha Mcclain	0
44	Networked systemic implementation	Ryan Frank	0
60	Robust eco-centric capacity	Brian Haney	0
127	Business-focused real-time benchmark	David Nelson	0
163	Object-based eco-centric challenge	Douglas Mccarthy	0
NULL	NULL	NULL	NULL

Inventory Insights

Query 10: Calculate the stock remaining after fulfilling all orders.

```
SELECT B.BookID,
        B.Title,
        B.Stock,
        COALESCE(SUM(O.Quantity),0) AS Order_Quantity,
B.Stock - COALESCE(SUM(O.Quantity),0) AS Remaining_Stock
FROM Books AS B
LEFT JOIN Orders AS O
ON B.BookID = O.BookID
GROUP BY B.BookID,B.Title,B.Stock
ORDER BY B.BookID ASC;
```

BookID	Title	Stock	Order_Quantity	Remaining_Stock
1	Configurable modular throughput	100	3	97
2	Persevering reciprocal knowledge user	19	0	19
3	Streamlined coherent initiative	27	5	22
4	Customizable 24hour product	8	0	8
5	Adaptive 5thgeneration encoding	16	8	8
6	Advanced encompassing implementation	2	0	2
7	Open-architected exuding structure	95	5	90
8	Persistent local encoding	84	3	81
9	Optimized interactive challenge	70	0	70
10	Ergonomic national hub	25	1	24
11	Secured zero tolerance time-frame	10	5	5
12	Polarized optimal array	63	0	63
13	Adaptive 5thgeneration orchestration	99	9	90
14	Re-engineered demand-driven parallelism	95	0	95
15	User-friendly motivating strategy	58	0	58

Key Insights and Recommendation

Insights

- 442 customers spent over $30, with higher activity in certain areas.
- "Kim Turner" is the highest-spending customer, contributing $1,398.90 in total purchases.
- 138 repeat customers, indicating strong retention potential.
- "Robust Tangible Hardware" was identified as the most frequently ordered book, highlighting its popularity among customers.
- Total revenue: $75,443.27

Recommendations

- Promote bestsellers and premium Fantasy titles strategically.
- Target high-value cities with localized promotions
- Ensure timely restocking of popular and out-of-stock titles to avoid missed sales opportunities.
- Implement a loyalty program for repeat customers to strengthen customer retention and drive repeat purchases, or create some exciting offers for top spenders.

Thank You so Much!